

Fixed-generator positive factorizations in Wajnryb's Artin A_5 quotient

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Abstract

We classify the positive words in the five standard generators of $B_6/\langle\langle ch^{-1} \rangle\rangle$ representing c^2 , where $c = (a_1 a_2 a_3 a_4)^5$ and $h = a_5 a_4 a_3 a_2 a_1^2 a_2 a_3 a_4 a_5$. Their lengths are 20, 30, 40, with respectively one, one, and two strict Hurwitz classes. Simultaneous conjugation identifies the two length-40 classes, giving the three representatives proposed in Wajnryb's question. The proof combines classical geometric and braid results with pure-braid linking constraints and an exhaustive, exact 810-word certificate for the length-20 case. The length spectrum and the twice-per-pair crossing theorem used at length 30 have prior sources; a reproducible six-strand certificate for the latter is supplied as additional evidence.

1 The question and equivalence conventions

Let $a_1, \dots, a_5$ be the standard generators of the braid group B_6 , with $a_i a_{i+1} a_i = a_{i+1} a_i a_{i+1}$ and $a_i a_j = a_j a_i$ for $|i - j| > 1$. Put

$$c = (a_1 a_2 a_3 a_4)^5, \quad h = a_5 a_4 a_3 a_2 a_1^2 a_2 a_3 a_4 a_5, \quad z = (a_1 a_2 a_3 a_4 a_5)^6, \quad G = B_6 / \langle\langle ch^{-1} \rangle\rangle.$$

The classical full-twist decomposition $z = ch = hc$ implies

$$c^2 = h^2 = z \quad \text{in } G; \quad z \text{ is central.} \tag{1}$$

In [7, Section 3, p. 126 of the linked online version], Wajnryb asks whether every positive word representing c^2 in G has length 20, 30, or 40, and is Hurwitz equivalent to one of the three displayed models h^2, z, c^2 .

Here a *positive word* means a literal word in these five fixed generators. Its factorization is the ordered tuple of its letters in G . A strict Hurwitz move replaces adjacent factors (x, y) by (xyx^{-1}, x) , or applies its inverse. We separately consider the equivalence generated by these moves and simultaneous conjugation of all factors by an element of G . Auroux introduces these operations separately in [1, Section 1, p. 131 of the linked online version]. Intermediate Hurwitz tuples may contain conjugates of generators; the inputs quantified here remain fixed-generator words.

Theorem 1. *The positive words in $a_1, \dots, a_5$ representing c^2 in G have precisely the following strict Hurwitz classes:*

length	number of classes	representatives
20	1	h^2
30	1	$(a_1 a_2 a_3 a_4 a_5)^6$
40	2	$(a_1 a_2 a_3 a_4)^{10}, \quad (a_2 a_3 a_4 a_5)^{10}.$

Every length-20 word in this set is literally one of the ten distinct cyclic rotations of h^2 . With simultaneous conjugation, there are exactly three classes, one at each length.

Thus the proposed three-model classification holds under the equivalence including simultaneous conjugation. Under strict moves alone the length-40 model splits into two classes. This theorem resolves the fixed-generator quotient question under both conventions. It does not classify arbitrary products of conjugates of generators or arbitrary nonseparating Dehn twists.

Prior results and contribution. The exact quotient is a classical genus-two, one-boundary mapping class presentation of Matsumoto [5, Theorem 1.3]: the relation $z = c^2$ is equivalent to $c = h$. The numerical spectrum 20, 30, 40 follows already from Sato’s geography theorem [6, Theorem 5.1, Table 1, Remark 5.1(1)] applied to the associated relatively minimal genus-two fibration with a section of square -1 and no reducible singular fibers. We use the later maximum-length theorem [2, Theorem A] and an elementary linking argument to obtain that spectrum here. Moreover, Elrifai–Morton [3, p. 496] report Casson’s general result that a positive braid with exactly two crossings per labelled pair is the full twist. They do not supply its proof or a separate Casson reference. Our six-strand computation verifies this known assertion in the required rank; it is not a new general theorem. The point of the note is the complete fixed-input orbit classification, especially the length-20 reduction and the strict versus simultaneous-conjugation distinction. A literature audit located no publication of this exact complete classification; that search does not establish absence of prior work.

2 Length and linking restrictions

2.1 Geometric bound

Take the standard five-curve chain on a genus-two surface with two boundary components and cap one boundary. The five twists define a homomorphism $B_6 \rightarrow \text{Mod}(\Sigma_{2,1})$. The odd and even chain relations send z and c^2 to the twist t_δ about the remaining boundary. Consequently $z = ch$ forces the images of c and h to agree, so the homomorphism factors through G . All five curves are nonseparating. The genus-two, one-boundary case of [2, Theorem A] says that the maximum number of nonseparating positive twists in a factorization of t_δ is 40. Hence every word under consideration has length $m \leq 40$. Only this homomorphism is needed for the bound, not an injectivity assertion for G .

The abelianization of B_6 is infinite cyclic, all a_i having the same class. The added relation has exponent sums 20 and 10, so $G_{\text{ab}} \cong \mathbb{Z}/10\mathbb{Z}$. The map sending each a_i to 1 modulo 10 proves that there are no additional abelian relations. Thus $m \in \{0, 10, 20, 30, 40\}$.

2.2 Normal-closure constraint

Let P_6 be the pure braid group. Both c and h are pure; the permutation map to S_6 therefore descends to G . A word representing c^2 is pure as a braid in B_6 . For a pure braid let ℓ_{ij} be half the signed crossing count of labelled strands i, j . These are integral homomorphisms $P_6 \rightarrow \mathbb{Z}$, and braid conjugation permutes their coordinates. For a positive pure word all $\ell_{ij} \geq 0$ and $m = 2 \sum_{i < j} \ell_{ij}$.

The linking vector β of c^2 equals 2 on pairs among $\{1, \dots, 5\}$ and 0 on pairs incident to 6. The linking vectors of conjugates of ch^{-1} are $(r_k)_{ij} = 1 - 2\mathbf{1}_{k \in \{i, j\}}$, $1 \leq k \leq 6$. Expressing $w(c^2)^{-1}$ as

a product of conjugates of the relator and its inverse gives integers $t_1, \dots, t_6$, of either sign, with

$$\ell_{ij}(w) = \beta_{ij} + T - 2(t_i + t_j) \geq 0, \quad T = \sum_{i=1}^6 t_i = m/10 - 4. \quad (2)$$

This is a necessary constraint only; no sufficiency of linking data is assumed.

Lemma 1. *Lengths 0 and 10 are impossible. At lengths 20, 30, 40, the linking vectors are respectively twice a star, the all-ones vector, and twice a complete graph on five vertices with one isolated vertex.*

Proof. Write $s = \sum_{i=1}^5 t_i$, so $t_6 = T - s$. The ten internal and five remaining inequalities in (2) are

$$t_i + t_j \leq \lfloor (T + 2)/2 \rfloor \quad (i, j \leq 5), \quad t_i + t_6 \leq \lfloor T/2 \rfloor \quad (i \leq 5).$$

For $T = -4, -3$ the internal sums yield $s \leq -3$; the remaining sums yield $s \geq \lceil (5T + 10)/4 \rceil = -2, -1$, respectively. This excludes $m = 0, 10$.

For $T = -2$, summing gives $-1 \leq s \leq 0$. If $s = -1$, then $t_6 = -1$, all $t_i \leq 0$ for $i \leq 5$, and exactly one is -1 . If $s = 0$, then $t_6 = -2$. Any positive maximum M among the first five entries would make the other four at most $-M$, giving $s \leq -3M < 0$. Thus the first five entries are all zero. Substitution gives precisely the six possible doubled stars.

For $T = -1$, the summed inequalities force $s = 0, t_6 = -1$; the remaining inequalities make the first five entries at most zero, so all are zero. Hence every $\ell_{ij} = 1$.

For $T = 0$, the remaining summed inequalities give $t_6 \leq 0$, so $s \geq 0$. If the maximum first entry $M \geq 1$, the other four are at most $1 - M$, giving $s \leq 4 - 3M \leq 1$; if $M \leq 0$, then $s \leq 0$. When $s = 0$, all entries vanish. When $s = 1, t_6 = -1$, each first entry is at most 1, at most one can be 1, and their sum forces one 1 and four zeros. These give exactly the six doubled five-vertex cliques. $\square$

The models in (1) realize all three permitted lengths. This proves the length assertion of Theorem 1.

3 Length 40: two strict classes

An isolated strand in the linking vector of a positive word has no crossings. It stays at its initial position k . If $2 \leq k \leq 5$, strands initially to its left and right cannot cross one another, contradicting the prescribed clique. Thus $k = 1$ or 6.

For $k = 6$, only $a_1, \dots, a_4$ occur. The four-chain map $B_5 \rightarrow \text{Mod}(\Sigma_{2,1})$ is injective by the A_4 case of Perron-Vannier, as stated in [7, Theorem 2.1]. Its chain neighborhood is the genus-two, one-boundary surface, up to a boundary collar. The geometric images of w and c^2 agree, so they are equal already in B_5 . The positive braid monoid embeds in B_5 [4, Section 4]; hence their words are connected by positive braid and commutation relations. A commutation is a Hurwitz swap. For $xyx = yxy$, two inverse Hurwitz moves give

$$(x, y, x) \mapsto (y, y^{-1}xy, x) \mapsto (y, x, x^{-1}y^{-1}xyx) = (y, x, y).$$

Thus every such tuple is strictly equivalent to c^2 . The other endpoint gives the class of $w_0 = (a_2a_3a_4a_5)^{10}$.

With $d = a_1a_2a_3a_4a_5$, braid relations give $da_id^{-1} = a_{i+1}$ for $i \leq 4$, so $dc^2d^{-1} = w_0 = z$ in G . The permutation images of the factor-generated subgroups for the two models are the S_5 fixing 6 and the S_5 fixing 1. A strict Hurwitz move preserves the subgroup generated by all factors, so these models are strictly inequivalent. Simultaneous conjugation by d identifies them.

4 Length 20: a complete finite certificate

Lemma 1 forces one labelled strand to cross each of the other five exactly four times; none of the other strands cross each other. Their relative order is constant. The distinguished strand makes a nearest-neighbor walk on six positions, returns to its start, and uses every edge four times. Crossing edge i records exactly the letter a_i . The fixed order of the other strands makes edge crossings correspond to crossings with individual labels. Conversely every such walk gives this pure-braid linking pattern.

Enumerate walks recursively, choosing either adjacent position whenever available and its edge has been used fewer than four times. At depth 20, retain a walk exactly when it returns to its starting position. The five edge caps force all edge counts to be four at that depth. This explores every candidate without any other pruning. The counts for starts $1, \dots, 6$ are

$$81, 162, 162, 162, 162, 81, \quad \text{total } 810.$$

We now give a presentation-compatible rejection test. Let $F_4 = \langle x_1, x_2, x_3, x_4 \rangle$. In the following table, a signed integer string is a free word: 12 denotes x_1x_2 , and $\bar{j}$ denotes x_j^{-1} . Use right actions, applying the first word and then the second.

	x_1	x_2	x_3	x_4
A_1	1	12	13	14
A_2	$\bar{1}\bar{2}1$	1	3	4
A_3	1	$2\bar{3}2$	2	4
A_4	1	2	$3\bar{4}3$	3
A_5	1	2	3	$3\bar{2}14$
A_1^{-1}	1	$\bar{1}2$	$\bar{1}3$	$\bar{1}4$
A_2^{-1}	2	$2\bar{1}2$	3	4
A_3^{-1}	1	3	$3\bar{2}3$	4
A_4^{-1}	1	2	4	$4\bar{3}4$
A_5^{-1}	1	2	3	$\bar{1}2\bar{3}4$

Substitution and free reduction verify both inverse compositions, the four adjacent braid relations, six distant commutations, and $A_c = A_h$. These finite word equalities prove that the tables define a right action of the presented group G . No faithfulness is asserted or required.

Apply this action to each of the 810 words and compare all four reduced images with those of c^2 . Exactly ten words pass; the passing set is literally the set of distinct cyclic rotations of h^2 . The numbers passing for the six starts are $1, 2, 2, 2, 2, 1$. These conclusions are exhaustive computation statements: the supplement contains the complete automorphism checks, all 810 word/image records, the survivor list, and independent implementations. All comparisons use full reduced words, not hashes.

Rejection is sound because the action is defined on G . Acceptance has a separate group certificate: $h^2 = z$ is central, so every cyclic rotation of h^2 has product z . For a tuple with central product, successive inverse Hurwitz moves moving its first factor to the end yield the ordinary cyclic rotation: the final conjugated factor is $z^{-1}g_1z = g_1$. Thus the ten words form exactly one strict Hurwitz class.

5 Length 30 and the reproducible crossing certificate

The linking vector is all ones, so every labelled pair crosses exactly twice. Casson's result, reported in [3, p. 496], would give $w = z$ in B_6 . For a checkable proof in the needed six-strand case, we

instead give a complete finite certificate of that equality.

Index the fifteen unordered label pairs lexicographically. A state consists of crossing counts $q_{ij} \in \{0, 1, 2\}$ and the order p of the six strand labels. Encode the counts by $q = \sum q_{ij} 3^{\text{index}(i,j)}$. Initially $q = 0$ and p is the identity. Each adjacent positive generator swaps the adjacent labels and increments their pair count if it is less than two. Counts increase by one, so the graph is a finite directed acyclic graph of depth at most 30.

The parity of each pair count determines whether that pair's relative order has reversed. Thus a reachable count vector determines a unique permutation, and merging by count code does not lose continuations. Breadth-first search exhausts all legal edges and yields 234,368 reachable states R and 711,342 outgoing edges. The full vector f , all counts two, is reachable with identity permutation. Complete candidate words are exactly paths from 0 to f .

A reachable vector q can continue to f exactly when $f - q$ is reachable. Indeed, reverse the sequence of swaps in a completing suffix, keeping generators positive: it starts at the identity and has labelled counts $f - q$. Conversely a reaching word for $f - q$ ends at the permutation of q , since the two vectors have the same parities. Reversing that sequence gives a legal completion from q ; nonnegative added counts never exceed the final counts. This is reversal of a positive word, not a braid inverse. Hence the coaccessible subgraph

$$S = \{q \in R : f - q \in R\}$$

has 90,921 states and 261,810 edges. Independent backward exploration from f gives the same set.

Use the faithful Artin action on F_6 , in the right-action convention

$$T_i(x_i) = x_i x_{i+1} x_i^{-1}, \quad T_i(x_{i+1}) = x_i, \quad T_i(x_j) = x_j \quad (j \neq i, i+1).$$

Faithfulness is classical [4, Section 1.6]; the inverse-generator and composition conventions change it by bijective braid automorphisms or anti-automorphisms, so still detect equality. Choose a reaching parent path to each state. A parent of a state in S lies in S , since it inherits a completion. Let $U(q)$ be the exact six reduced free-word images along this path. The certificate verifies, on *every* edge $q \xrightarrow{i} q'$ in S ,

$$T_i(U(q)) = U(q') \tag{3}$$

by full word equality, and verifies that $U(f)$ equals the action of the displayed word z . Induction along an arbitrary complete path now gives its action as $U(f)$. Faithfulness proves $w = z$ in B_6 . Positive monoid embedding and the Hurwitz realization of braid relations then put all length-30 words in one strict class.

Counting states or equating hashes would not prove (3). Python and C++ implementations perform all 261,810 literal edge comparisons. The distributed controls `independent_20_30.py` and `independent_ranks1_6.py` each reconstruct the backward graph and reproduce them. Their complete sorted 90,921-record action streams agree byte for byte. For integrity, the record stream has 15,258,431 bytes and SHA-256

af2b8ec569d613e4f3d8ba3b72d18e85e8c6
125f4265057b7f4c485d544d159bf

Equal pair counts alone do not determine a braid: the positive prefixes $a_1^2 a_2^2$ and $a_2^2 a_1^2$ have equal pair counts and permutations but different free-group actions. They lie outside S , and the independent negative controls verify this distinction.

6 Verification, scope, and disclosure

The supplement distributes the original four-turn proofs and checkers, exact receipts, complete compressed action records, independent finite controls, and a scope-specific priority report. The length-20 test requires only Python’s standard library; the second large certificate additionally uses a C++17 compiler. No external group solver or numerical approximation is needed. Source hashes and package hashes are integrity checks; the completeness arguments and exact comparisons above remain the mathematical content of the certificates. The geometric length theorem, the chain embedding, faithful Artin action, and positive monoid embedding are credited classical inputs rather than claims proved by these programs.

Different lengths cannot be Hurwitz equivalent. At length 40 the generated-subgroup invariant proves that the two strict classes are distinct; the explicit conjugation merges them. The other two lengths each have one strict class. This completes the proof of Theorem 1.

Use of AI and review status. AI tools were used extensively in developing the proofs, writing and running verification code, searching the literature, drafting this note, and conducting adversarial reviews. Independent AI review paths are recorded in the accompanying research audit. This manuscript is an unrefereed preprint; it has not received external human peer review. The stated result concerns only the fixed standard-generator input universe and both expressly defined equivalence conventions.

References

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